CGSNet: Category Prior-Guided Self-Supervised Semantic Segmentation for Remote Sensing Images

Jiahang Liu, *Senior Member, IEEE*, Jian Cui, Mao Guo, Weijian Chi, Yue Ni, Xifu Sun, and Ji Luan

***Abstract*—Multimodal fusion methods have shown great potential in remote sensing image analysis, but existing approaches rely heavily on massive amounts of annotated data. This is not only costly and time-consuming but also prone to subjective bias. To address this issue, we propose a category-prior-based self-supervised framework, CGSNet, which uses category prior maps extracted from multispectral images as supervisory signals for end-to-end training. An adaptive confidence-weighted pseudo-label generation mechanism is designed to alleviate noise and errors in prior maps by replacing binary labels with continuous confidence maps, enabling the learning of uncertain interclass features. In addition, a multispectral feature-guided refinement strategy utilizes color and texture information to calibrate class transition regions and enhance the discriminative power of pseudo-labels in complex scenes. A dynamic mask selection strategy further enhances the model's robustness and generalization capabilities through progressive learning. Experiments demonstrate that CGSNet achieves state-of-the-art performance without the need for human annotation, achieving an Mean Intersection over Union (mIoU) score of 78.46% on the Gaofen image dataset (GID) (vegetation) dataset and 79.58% on the Zurich (vegetation) dataset—12.22% and 15.07% higher than existing methods, respectively—while exhibiting strong cross-dataset zero-shot generalization capabilities. The code will be available at https://github.com/NUAALISILab**

***Index Terms*—Category prior, remote sensing, self-supervised, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing images, also referred to as land-cover classification, aims to achieve pixel-level localization of objects and assign each pixel in the image to a specific category [1]. This technology plays a vital role in various remote sensing applications [2], [3], [4], [5], [6].

Benefiting from powerful multilevel feature learning capabilities, deep learning methods have demonstrated remarkable performance in remote sensing image semantic segmentation. However, significant intraclass variability of the same land-cover type across different images, coupled with high interclass similarity between different categories, often hinders models from learning unified and robust feature representations, thereby limiting their ability to effectively extract and utilize discriminative features of land-cover classes [7]. Multimodal data can provide richer information from diverse perspectives, showing considerable potential for complex scenarios and high-precision segmentation. Fusing multimodal data for semantic segmentation of remote sensing images enables the full exploitation of complementary characteristics from different data types, thereby acquiring more comprehensive and abundant semantic information and improving the accuracy and reliability of segmentation [8].

However, as a data-driven paradigm, deep learning-based multimodal fusion methods require a large number of annotated samples for training, and their performance is heavily dependent on the quantity and quality of the training data [9]. As manual annotation is time-consuming and labor-intensive, constructing large-scale, high-quality annotated datasets remains challenging [10]. Moreover, as shown in Fig. 1, the complexity of object boundaries in remote sensing images makes the annotation process susceptible to subjective bias, leading to potential label noise and dataset contamination.

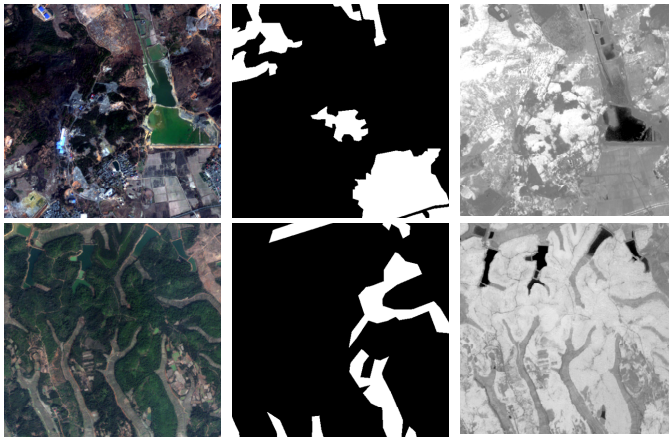


Fig. 1. Vegetation labels of the GID dataset and corresponding NDVI.

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To alleviate the scarcity of annotated data, numerous studies have focused on few-shot learning methods [11]. Semi-supervised learning leverages both labeled and unlabeled samples for land-cover classification, among which graph-based learning represents a typical semi-supervised paradigm for remote sensing data [12]. Graph convolutional networks (GCNs) construct graph structures from labeled and unlabeled samples and extract deep graph features for classification [13]. As a data augmentation technique, generative adversarial networks (GANs) utilize both real annotated samples and generated samples for semi-supervised classification [14]. However, due to the inherent complexity of remote sensing data, these methods still face difficulties in generating realistic samples and constructing effective graph structures. Another research direction is weak supervision. For example, Siamese networks are employed to extract high-dimensional nonlinear spatial–spectral features from a limited number of annotated samples. Few-shot learning typically trains a feature extractor on a source domain and directly transfers it to a target domain for feature extraction; the extracted features are then combined with a small number of labeled samples to perform classification [15]. Transfer learning mainly consists of two stages: pretraining and fine-tuning. In the pretraining stage, low- and mid-level features are learned from a source dataset, and then the network is adapted using a small number of annotated samples from the target domain for the target classification task [16]. To address the domain shift problem, domain adaptation methods aim to minimize the discrepancy between the source and target domains, thereby reducing statistical distribution misalignment [17].

Although significant progress has been made with the above methods, existing few-shot learning models still rely on a limited amount of annotated data for pretraining or fine-tuning, making them unsuitable for fully unlabeled tasks. Furthermore, current approaches fail to effectively encode the specific reflection and absorption mechanisms of land-cover objects (e.g., the high near-infrared (NIR) reflectance of vegetation) into the feature learning pipeline. The lack of modeling and utilization of the spectral physical properties of ground objects results in limited model robustness and generalization capability.

To address these challenges, this article proposes a self-supervised learning (SSL) framework for semantic segmentation of remote sensing images based on category prior guidance (CGSNet). This framework creates an adaptive confidence-weighted pseudo-label generation (APL) mechanism, utilizing multispectral images and their derived category prior information to generate high-quality pseudo-labels. It also constructs a multimodal cross-gated self-attention module to achieve complementary fusion of cross-modal features. Unlike existing self-supervised methods, CGSNet utilizes category prior information automatically extracted from multispectral remote sensing images as a supervisory signal, establishing an end-to-end SSL paradigm that achieves semantic segmentation of remote sensing images without any manually labeled data. The main contributions of this work are summarized as follows.

1) A category prior-guided self-supervised semantic segmentation framework, termed CGSNet, is proposed. It pioneers the use of category prior information derived from multispectral remote sensing images as supervisory signals, establishing a novel SSL paradigm independent of external supervision, thereby achieving semantic segmentation of remote sensing imagery without any manual annotations.
2) An APL mechanism is introduced, where continuous confidence maps replace traditional binary labels to effectively characterize transitional features in interclass uncertain regions. A multispectral feature-guided optimization strategy is further incorporated, leveraging the rich color and texture information in multispectral images to dynamically refine category transition areas in pseudo-labels. In addition, a dynamic mask selection strategy is integrated to enable progressive learning, enhancing the model's robustness to challenging samples and its adaptability across diverse scenarios.
3) A multimodal cross-gating self-attention module (CGSM) is introduced to adaptively fuse multispectral and prior knowledge through a gating mechanism, enabling explicit learning of spectral-physical properties for improved feature representation and semantic consistency.

Experimental results on different datasets demonstrate that the proposed CGSNet exhibits strong robustness and generalization capability across various tasks. Compared to other state-of-the-art methods, CGSNet effectively leverages multispectral image data and category prior information without any manual annotations, generating high-quality pseudo-labels and achieving superior detection accuracy in vegetation, water body, and shadow detection tasks.

The remainder of this article is organized as follows. Section II first reviews the related works on remote sensing segmentation and self-supervised semantic segmentation methods. After that, Section III presents the structure of the proposed CGSNet, whereas Section IV provides details on the extensive experiments conducted. Finally, the conclusion is given in Section V.

II. Related Works

A. Traditional Remote Sensing Image Segmentation

Classical remote sensing image segmentation methods primarily rely on spectral, spatial, or textural features, employing manually designed feature extraction and segmentation strategies. Common approaches include threshold-based, edge detection-based, and region-based segmentation methods.

Threshold-based methods classify pixels by comparing their values with predefined thresholds [18]. Zhang and Li [19] proposed a minimum class mean absolute deviation (MCMAD)-based algorithm, which enhances segmentation efficiency and accuracy for massive remote sensing data with complex textures by reducing 2-D histograms to 1-D and integrating spectral information. In addition, spectral indices such as the normalized difference vegetation index (NDVI)

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and normalized difference water index (NDWI) are widely used for specific feature extraction. Liu et al. [20] introduced a shadow detection method leveraging multispectral physical properties, which mitigates misclassification between shadows and dark features (e.g., vegetation and water bodies) through transformation models suppressing nonshadow objects and combining color difference composite index (CDCI) with color purity index (CPI). Although threshold-based methods are computationally efficient and straightforward, they are highly sensitive to noise and outliers, often exhibiting significant performance degradation under conditions of low pixel contrast, complex backgrounds, or variable illumination.

In addition, some methods identify abrupt edge features by calculating gradients using spatial filters and leverage these edge features to achieve image segmentation [21], [22]. For instance, Bhatti et al. [23] developed a quaternion and Clifford algebra-based edge detection approach, which improves edge recognition accuracy for small objects in satellite imagery by employing multidirectional frequency filtering and quaternion Fourier transform (QFT), addressing challenges posed by noise and spectral overlap. While effective for distinct boundaries, these methods struggle with noise robustness and edge precision.

Region-based methods merge homogeneous areas based on similarity metrics. Common unsupervised algorithms include K-means, hierarchical clustering, and fuzzy C-Means (FCMs), making them suitable for large-scale unlabeled remote sensing data. Fan et al. [24] proposed a region-merging method integrating texture pattern similarity measure (TPSM) and statistical similarity, resolving false mergers of texturally similar regions in synthetic aperture radar (SAR) images. Wang et al. [25] introduced a hybrid segmentation framework balancing inter-region heterogeneity and intraregion homogeneity (OHRH), improving geo-object delineation by addressing inconsistencies in traditional region-merging algorithms.

Traditional approaches, relying on spectral, spatial, and textural features, have achieved notable progress in simple scenarios. However, their dependence on handcrafted features limits robustness under variable illumination or complex terrains, often resulting in blurred boundaries and accuracy loss. Moreover, shallow statistical features fail to capture high-order patterns, leading to unstable performance and poor scalability for large-scale automated processing.

B. Deep Learning-Based Remote Sensing Image Segmentation

Deep convolutional neural networks (CNNs) excel in semantic segmentation of high-resolution remote sensing images (HRRSIs), which are characterized by large object-size variations, complex distributions, and ambiguous contours. Zhang et al. [26] aggregated multiscale contextual information via high- and low-resolution branches in HRNet. Dong et al. [27] designed DenseUNet for urban small-object segmentation by cascading shallow detail features and deep semantic features. Ma et al. [28] proposed a foreground activation-driven framework for HRRSIs, while Li et al. [29] exploited multiattention modules to capture context dependencies. To address edge inaccuracy, Ni et al. [8] developed an edge-guided network with oriented convolution and spatial-semantic feature aggregation, enhancing segmentation precision by integrating directional features.

Despite transformers' success, their high computational costs hinder large-scale applications. Hybrid CNN-Transformer models dominate remote sensing tasks. Cui et al. [30] proposed a global context dependency-aware network (GCDNet), combining dot-product attention and edge-aware optimization to overcome limited receptive fields and imprecise edge localization. Ni et al. [31] further introduced a class-guided global–local feature interaction network (CGGLNet), mitigating intraclass variation and interclass aliasing through category-guided context modeling.

While deep learning has significantly advanced segmentation accuracy, its performance heavily relies on large annotated datasets. Labeling noise due to subjective biases and complex boundaries further degrades model generalizability. Under limited labeled data, these methods often exhibit substantial performance drops, restricting real-world applicability.

C. Self-Supervised Semantic Segmentation

The SSL paradigm utilizes the intrinsic attributes of data to learn representative features, and learned characteristics are employed for downstream recognition tasks. Aiming at semantic segmentation, many self-supervised dense representation learning works introduce domain-specific priors, including contour detectors, saliency detectors, or clustering.

To address the challenges of unsupervised object localization and segmentation in complex multiobject scenes, Melas-Kyriazi et al. [32] propose a novel spectral segmentation approach that reformulates image decomposition as a graph partitioning problem through Laplacian eigenanalysis of self-supervised feature affinity matrices, achieving superior performance over state-of-the-art methods on Pascal VOC and MS-COCO datasets while enabling practical image editing applications. To address the challenge of semantic inconsistency in pixel-level clustering for unsupervised semantic segmentation, Li et al. [33] proposed an adaptive conceptualization approach (ACSeg) featuring learnable prototypes and an adaptive concept generator, which achieved state-of-the-art performance by dynamically adapting to varying image complexities through modularity loss optimization.

While these methods can reduce dependence on annotated data, the models still rely on external supervision and remain inapplicable in fully unannotated scenarios.

III. Methodology

A. Framework Overview

This article proposes a category prior-guided self-supervised semantic segmentation framework for remote sensing images (CGSNet). The framework addresses the dependency on annotated data in data-driven methods by designing a confidence-aware pseudo-label generation mechanism and a cross-modal complementary feature fusion module, which fully utilizes the texture features of ground objects and the spectral characteristics of category priors. As shown in Fig. 2(a), CGSNet

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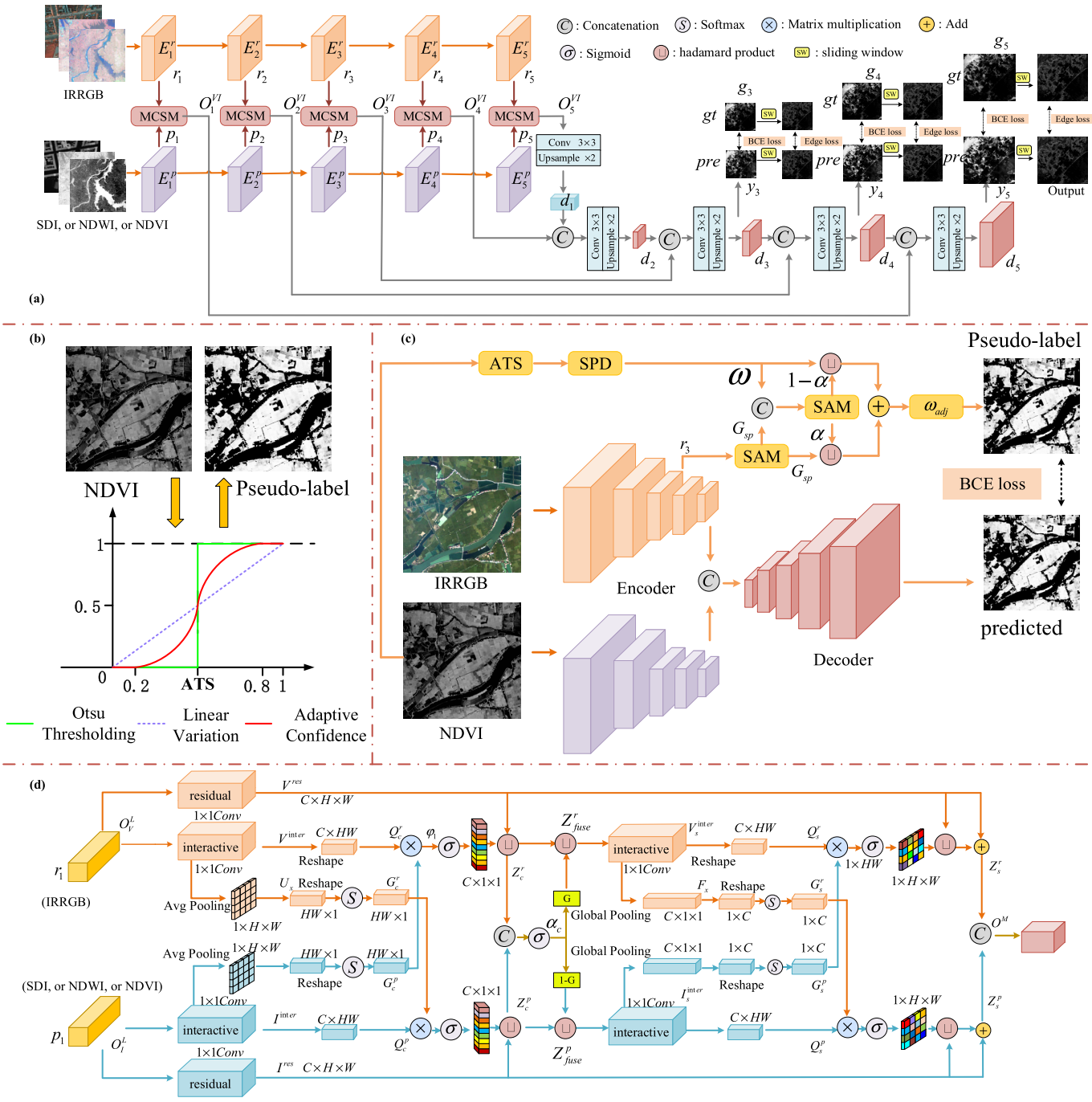


Fig. 2. (a) Overview of CGSNet for semantic segmentation. The inputs are a multispectral image and another modality of data (e.g., SDI, NDWI, and NDVI). (b) Smooth probability distribution. (c) APL. (d) Detailed architecture of the multimodal CGSM.

adopts an encoder–decoder architecture with MobileNetV2 as the backbone network. An adaptive confidence-weighted label generation mechanism that produces training supervision labels from spectral indices computed from remote sensing data. A dual-branch network structure in the encoder that extracts features from multispectral images I_r NIR, red, green, blue) and category prior images I_p (e.g., NDVI). Each branch outputs feature maps at five different scales, denoted as r_a and p_a, where $a \in \{1, 2, 3, 4, 5\}$. Besides, a cross-modal complementary feature fusion module that extracts complementary features across modalities in both channel and spatial dimensions. The decoder generates outputs at five different scales, denoted as d_a ($a \in \{1, 2, 3, 4, 5\}$). A boundary refinement algorithm is incorporated into the decoder outputs d_3, d_4, d_5 to enhance edge-aware learning capabilities.

B. Adaptive Confidence-Weighted Pseudo-Label Generation

Traditional threshold-based segmentation methods directly binarize category prior images using fixed thresholds, ignoring uncertainties in transition regions. As shown in Fig. 2(c), to address this limitation, we propose an APL mechanism that

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leverages the complementary characteristics of multimodal remote sensing data. This mechanism generates high-quality pseudo-labels through adaptive threshold segmentation and multispectral image feature guidance.

1) Adaptive Threshold Selection: To address the nonuniform intraclass distribution in remote sensing images caused by varying illumination conditions, seasonal changes, and differences in ground cover, this section proposes an adaptive threshold selection (ATS) method based on weighted interclass variance. The approach aims to derive a more rational segmentation threshold by leveraging prior category information. It specifically overcomes a limitation of the traditional Otsu algorithm, in which the selected threshold tends to deviate when processing histograms with asymmetric distributions. By incorporating the skewness statistic, which characterizes distribution asymmetry, the calculation of interclass variance between foreground and background is adaptively weighted, thereby enhancing the robustness of threshold selection under imbalanced proportions of target and background pixels.

For the category prior image $V \in \mathrm{R}^{H\times W}$ (where H, W denote the height and width of the image), the pixel values are defined as $v \in [0, 1]$. First, the normalized grayscale histogram $p(v)$ of the image is computed. Based on this histogram, the skewness coefficient s is calculated to quantify the degree of asymmetry in the histogram distribution

$$\begin{cases} \mu = \displaystyle\int_0^1 v \cdot p(v)dv \\ \theta = \sqrt{\displaystyle\int_0^1 (v-\mu)^2 \cdot p(v)dv} \\ s = \dfrac{E\left[(v-\mu)^3\right]}{\theta^3} \end{cases} \tag{1}$$

where μ represents the global mean, θ is the standard deviation, and E denotes the expectation operator. The skewness coefficient s reflects the distribution characteristics of the target class. $s > 0$ indicates the presence of a small number of high-brightness pixels (foreground), while $s < 0$ suggests a concentration of low-brightness pixels (background). To integrate skewness information into the threshold decision process, a normalized weight adjustment coefficient β is constructed to reflect the degree of distribution asymmetry

$$\beta = \frac{|s|}{|s|+1}, \quad \beta \in [0, 1). \tag{2}$$

Then, a novel threshold selection criterion based on weighted interclass variance is formulated. The weighted interclass variance $\theta_\omega^2(t)$ is defined as

$$\begin{aligned} \theta_\omega^2(t) = &\left[\omega_0(t) \cdot (1 + \beta \cdot sign(s))\right] \\ &\cdot \left[\omega_1(t) \cdot (1 - \beta \cdot sign(s))\right] \\ &\cdot \left[\mu_0(t) - \mu_1(t)\right]^2 \end{aligned} \tag{3}$$

where $\omega_0(t)$ and $\omega_1(t)$ denote the proportions of foreground and background pixels, respectively, $\mu_0(t)$ and $\mu_1(t)$ represent the mean grayscale values of the foreground and background classes, and sign (s) is the sign function

$$sign(s) = \begin{cases} 1, & s > 0 \\ -1, & s < 0 \\ 0, & s = 0. \end{cases} \tag{4}$$

This weighting mechanism adaptively adjusts the class weights according to the skew direction of the histogram. When $s > 0$, the weight for the foreground class is increased, thereby raising the threshold. Conversely, when $s < 0$, the background class is emphasized, leading to a lower threshold. The optimal initial segmentation threshold T is obtained by maximizing the weighted interclass variance

$$T = \arg\max \theta_\omega^2(t), \quad t \in [0, 1]. \tag{5}$$

To mitigate misclassification in transitional areas caused by the abrupt binary partitioning of traditional fixed-threshold methods, the category prior image is subsequently divided into three regions based on T: the background region V_b, the transition region V_t, and the foreground region V_p. Pseudo-labels are then generated adaptively to refine segmentation boundaries according to the spectral heterogeneity of different geographical areas. The region-wise label assignment is defined as

$$\begin{cases} \text{label}(V_b) = 0, & v < T - \tau \\ \text{label}(V_t) = \omega_{adj}, (\omega_{adj} \in (0, 1)), & T - \tau \le v \le T + \tau \\ \text{label}(V_p) = 1, & v > T + \tau \end{cases} \tag{6}$$

where τ denotes the transition width, set to 0.2 in this study.

2) Confidence-Weighted Pseudo-Label Generation:

a) Smooth probability distribution: As shown by the red line in Fig. 2(b), we generate pseudo-labels for the transition region V_t using a confidence-weighted approach based on a Sigmoid function soft threshold mapping

$$\omega_{(x,y)} = \frac{1}{1 + e^{\left(-\frac{V_{(x,y)} - T}{\tau}\right)}} \tag{7}$$

where $V_{(x,y)}$ denotes the pixel value of the category prior image at position (x, y). This design ensures that the pseudo-label distribution satisfies $\omega_{(x,y)} \to 0$ when $V_{(x,y)} \ll T$, and $\omega_{(x,y)} \to 1$ when $V_{(x,y)} \gg T$. Within the transition region [T-τ, $T + \tau$], the weights exhibit a smooth probability transition, enabling the model to learn uncertain features and effectively mitigate false detections caused by fixed thresholds.

b) Multispectral image feature guidance: To fully utilize the spatial texture information of multispectral images, we design a multispectral image feature guidance module to further optimize the labels in transition regions and enhance the discriminative ability of pseudo-labels in complex areas. The multispectral image features $r_3 \in \mathrm{R}^{C\times H'\times W'}$ (where C is the number of channels) extracted by the encoder are used as guiding features. A spatial attention module (SAM) generates the spatial guidance feature map G_{sp}

$$G_{sp} = \sigma\left(Conv_{1\times1}\left(ReLU\left(BN\left(Conv_{3\times3}(r_3)\right)\right)\right)\right) \tag{8}$$

where $Conv_{3\times3}$ and $Conv_{1\times1}$ denote 3×3 and 1×1 convolutions, respectively. BN represents batch normalization, ReLU

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is the activation function, and σ is the Sigmoid function that normalizes the output to [0, 1].

The transition region confidence image features ω and the multispectral guidance feature map G_{sp} are concatenated, and an adaptive weight is computed through a feature fusion attention module. First, the two feature maps are concatenated along the channel dimension, and a SAM computes the attention weight map α

$$\alpha = \sigma\left(Conv_{1\times1}\left(ReLU\left(BN\left(Conv_{3\times3}\left(Concat\left(\omega, G_{sp}\right)\right)\right)\right)\right)\right). \tag{9}$$

The final adjusted confidence map is

$$\omega_{adj} = \alpha \cdot G_{sp} + (1-\alpha)\cdot\omega. \tag{10}$$

This method enables the model to adaptively learn the fusion weights of multispectral image features and category prior features, leveraging the texture information of multispectral images to improve pseudo-label accuracy in complex transition regions.

3) Morphological Optimization: Confidence map ω_{adj} often contains isolated noise and small holes. We employ morphological operations for optimization, including opening and closing operations, as well as area filtering.

The opening operation (erosion followed by dilation) effectively removes isolated noise points while preserving main structural features. It is calculated as

$$B_{open} = D\left(Er\left(\omega_{adj}\right)\right) \tag{11}$$

where $Er(\cdot)$ denotes the erosion operator

$$Er\left(\omega_{adj}\right)(x,y) = \min_{(i,j)\in \mathrm{N}} \omega_{adj}(x+i, y+j) \tag{12}$$

where N represents the neighborhood range (set to 4×4 in this article). $D(\cdot)$ denotes the dilation operator

$$D\left(\omega_{adj}\right)(x,y) = \max_{(i,j)\in \mathrm{N}} \omega_{adj}(x+i, y+j). \tag{13}$$

The closing operation (dilation followed by erosion) effectively fills small holes and enhances regional connectivity while maintaining structural integrity

$$B_{close} = Er\left(D\left(B_{open}\right)\right). \tag{14}$$

The morphologically processed binary map B_{close} exhibits enhanced structural integrity and smoother edges. The final pseudo-label L_{pseudo} is generated by fusing the morphological processing result with the confidence weight

$$L_{pseudo} = B_{close} \odot \omega_{adj} \tag{15}$$

where $\odot$ denotes the Hadamard product.

Morphological processing further suppresses small-scale noise, while confidence weighting provides a pixel-level reliability measure. The generated pseudo-labels maintain the integrity of main detection regions and exhibit a continuous probability distribution in transition regions, providing fine-grained supervision signals for model training. The confidence weighting mechanism essentially constructs an adaptive attention map, guiding the model to focus on high-confidence regions during learning while capturing spectral–spatial feature correlations in transition regions through gradient propagation. This soft supervision strategy effectively mitigates the accumulation of erroneous labels and establishes a robust foundation for SSL.

4) PMG Strategy: To effectively utilize pseudo-labels in self-supervised training and improve model learning efficiency, we propose a progressive mask generation (PMG) strategy. This method dynamically adjusts the selection of masked regions, implementing a learning process from hard to easy samples.

a) Sample difficulty awareness: A difficulty map D_{map} is generated based on the pseudo-label map

$$D_{map} = 1 - \left|L_{pseudo} - 0.5\right| \times 2. \tag{16}$$

Regions with confidence close to 0.5 correspond to great difficulty.

b) Progressive mask generation: The masking strategy is adjusted according to the training progress to achieve learning from hard to easy samples. n represents the current training epoch, and N represents the total number of epochs. Define $L = n/N$. When $L < 0.3$, train mask hard regions ($D_{map} > 0.7$). When $0.3 \le L \le 0.7$, train mask medium-difficulty regions ($0.3 \le D_{map} < 0.7$). When $L > 0.7$, train mask easy regions ($D_{map} < 0.3$).

This dynamically adjusted masking ratio avoids the training instability caused by fixed masking strategies. Moreover, the progressive masking strategy enables the model to learn hard samples in the early stages and gradually transition to easy samples, allowing the model to progressively build a multilevel feature understanding from details to the whole, effectively improving robustness and generalization in complex scenes.

C. Multimodal CGSM

To effectively integrate features from multispectral and prior data while enhancing cross-modal interaction, we propose a multimodal CGSM. This module employs a lightweight polarized self-attention mechanism to capture long-range dependencies in both channel and spatial dimensions, and a gated fusion strategy to adaptively combine complementary information.

As illustrated in Fig. 2(d), the module processes each modality through a dual-path structure. We take the multispectral image path for illustration. We apply two parallel 1×1 convolutions to decompose it into a residual vector V^{res} and an interaction vector V^{inter}. The interaction vectors are used to compute cross-modal attention, while the residual vectors preserve original features.

In the channel-wise cross-attention stage, we compute attention weights by pooling and projecting the interaction vector along the channel dimension, followed by Softmax normalization

$$G_c^r = Soft\max\left(Avgpool_{ch}\left(V^{inter}\right)\right). \tag{17}$$

A query vector Q_c^r is derived from the same interaction vector via a linear projection $\mu(\cdot): \mathrm{R}^{C\times H\times W} \to \mathrm{R}^{C\times HW}$. The prior path and the multispectral path adopt the same network

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structure, and the vectors Q_c^p and G_c^p of the prior path are obtained using the same method.

The similarity matrix between the query vector and the attention map is computed and activated via a sigmoid function. The output is obtained by weighting the residual vector

$$Z_c^r = \sigma\left(Q_c^r \otimes G_c^p\right) \odot V^{res} \tag{18}$$

$$Z_c^p = \sigma\left(Q_c^p \otimes G_c^r\right) \odot I^{res}. \tag{19}$$

The same process is applied to both modalities, yielding channel-refined features Z_r^c and Z_p^c.

A gating fusion mechanism is then applied to adaptively combine the two modalities. The features are concatenated and passed through a convolutional layer followed by a sigmoid function to generate a gating weight map $\alpha_c \in [0, 1]$:

$$\alpha_c = \sigma\left(Conv\left(Concat\left(Z_c^r, Z_c^p\right)\right)\right). \tag{20}$$

The fused output is computed as

$$Z_{fuse}^r = \alpha_c \odot Z_r^c \tag{21}$$

$$Z_{fuse}^p = (1 - \alpha_c) \odot Z_p^c. \tag{22}$$

In the spatial cross-attention stage, a similar procedure is applied to the fused feature Z_{fuse}^r and Z_{fuse}^p, but with pooling and projection performed along the spatial dimension. The resulting spatial attention map is used to modulate the residual features, enhancing salient spatial structures. The spatial cross-attention is computed as

$$G_s^r = Soft\max\left(\xi\left(Avgpool_{sp}\left(V_s^{inter}\right)\right)\right) \tag{23}$$

$$G_s^p = Soft\max\left(\xi\left(Avgpool_{sp}\left(I_s^{inter}\right)\right)\right) \tag{24}$$

$$Z_s^r = \sigma\left(Q_s^r \otimes G_s^p\right) \odot V^{res} + V^{res} \tag{25}$$

$$Z_s^p = \sigma\left(Q_s^p \otimes G_s^r\right) \odot I^{res} + I^{res} \tag{26}$$

$$O^M = Concat\left(Z_s^r, Z_s^p\right) \tag{27}$$

where $\xi(\cdot)$ denotes $\xi(\cdot) : \mathrm{R}^{C\times1\times1} \rightarrow \mathrm{R}^{1\times C}$ a projection function. The final output O^M is obtained by concatenating the spatially enhanced features from both paths.

This design enables efficient and selective fusion of multimodal features, emphasizing complementary information while suppressing redundancy, with minimal computational overhead. The combination of cross-modal attention and gating mechanisms allows the model to dynamically prioritize features from different modalities, improving discriminability in semantic segmentation.

D. Multiscale Edge Optimization

Unlike traditional methods that learn edge features from feature maps, segmentation prediction maps contain richer boundary information that can be directly optimized. We integrate independent edge loss calculation into the model. The boundary refinement module consists of a cross-entropy loss function L_{BCE} and an edge optimization loss function L_{edge}

$$L_{BOA} = L_{BCE}(g, y) + L_{\text{edge}}\left(g_{\text{edge}}, y_{\text{edge}}\right) \tag{28}$$

where g represents the ground truth, and y represents the prediction results. The boundary labels g_{edge} are generated from the label map using a sliding window. Similarly, a lightweight edge extractor based on a sliding window is applied to the top of the segmentation network to directly extract boundary information y_{edge} from the prediction map.

To minimize computational overhead, the edge extractor employs a simple sliding window operation that computes the difference between the maximum and minimum values within the window, effectively filtering out pixels similar to neighborhood values and retaining pixels with significant changes to separate edge regions. During training, prediction maps y_3, y_4 and y_5 are obtained from feature maps d_3, d_4 and d_5, with corresponding ground truths denoted as g_3, g_4 and g_5. During inference, only the final output is used as the model prediction. The overall loss function is expressed as

$$\begin{aligned} L_{BOA} = {} & L_{BCE}(g_3, y_3) + L_{BCE}(g_4, y_4) + L_{BCE}(g_5, y_5) \\ & + L_{\text{edge}}\left(g_{3\text{edge}}, y_{3\text{edge}}\right) + L_{\text{edge}}\left(g_{4\text{edge}}, y_{4\text{edge}}\right) \\ & + L_{\text{edge}}\left(g_{5\text{edge}}, y_{5\text{edge}}\right). \end{aligned} \tag{29}$$

The multiscale edge optimization (MEO) module directly extracts edge information from pixel-level segmentation predictions, mitigating information loss in low-dimensional features and enhancing edge semantic features without introducing additional parameters or computational cost.

Unlike existing semantic segmentation networks that primarily improve feature extraction capabilities by optimizing network structures, our model adopts a simple network architecture to directly learn prior knowledge from the spectral attribute information embedded in images. By leveraging the pixel-level category probability distribution in spectral indices, the deep neural network is guided to learn category-specific intrinsic correlation features from multispectral images. This method combines traditional model-driven approaches with data-driven deep neural network models, integrating physical and spectral features as category priors with deep learning to enhance object detection capabilities in complex scenes and improve adaptability to diverse environments.

IV. Comparison and Discussion

In this section, we conduct self-supervised semantic segmentation experiments for water bodies, vegetation, and shadows in remote sensing images based on different data source combinations. Subsequently, ablation studies are performed on the individual modules of the proposed method. Finally, the effectiveness and generalization capability of the proposed method are further validated through comparisons with existing methods.

A. Dataset

We conducted experiments on water body, vegetation, and shadow detection using different satellite remote sensing datasets, including the Gaofen image dataset (GID), Zurich dataset, and GF-2 dataset.

1) *Gaofen Image Dataset:* The GID is an open-source satellite remote sensing dataset containing 150 annotated land cover images. All images were captured by the GF-2 satellite with a spatial resolution of 4 m. Each image contains four bands: red (R), green (G),

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blue (B), and NIR. We obtained NDVI and NDWI data corresponding to the multispectral images through band math. Following the data processing method in the literature [34], we redefined the labels of the original GID dataset to construct the GID (water) dataset and the GID (vegetation) dataset, which were used for supervised model training in comparative experiments. Given the poor annotation quality of the vegetation category in the original GID dataset, we created fine vegetation masks for the test set through manual annotation to correct erroneous annotations, enabling more reliable performance evaluation.

2) *Zurich Dataset:* The Zurich dataset contains 20 very high-resolution (VHR) multispectral images acquired by the QuickBird satellite over Zurich, Switzerland, with a spatial resolution of 0.61 m. Each image also contains R, G, B, and NIR bands, and corresponding NDVI and NDWI data were obtained through band math. For the water body detection experiments, we redefined the original dataset by categorizing rivers as water and other classes as background, combining this with NDWI to construct the Zurich (water) dataset. For the vegetation detection experiments, forests and grasslands were classified as vegetation, with others as background, combined with NDWI to construct the Zurich (vegetation) dataset. The redefined labels were used for supervised model training in comparative experiments. We also manually annotated finer vegetation masks for the Zurich (vegetation) dataset to serve as test set labels.
3) *GF-2 Dataset:* We selected high-resolution GF-2 remote sensing images covering urban areas for shadow detection experiments. The original image size is 7300×6908 pixels, containing blue, green, red, and NIR bands. For convenient model training, they were cropped into 256×256 pixel patches. The shadow detection index (SDI) [20] was adopted as prior information to guide the model for shadow detection.

B. Implementation Details

The PyTorch library was used to implement and train CGSNet, which was trained using Adam optimization, and we set the learning rate of our training phase to $1e^{-5}$. The Mobilenetv2 backbone network was initialized with pretrained weights from the ImageNet dataset, and the other parts were initialized randomly.

C. Evaluation Metrics

The evaluation of experimental outcomes was performed using metrics including overall accuracy (OA), $F1$-score($F1$), and mean intersection over union (mIoU). The calculation formulas for these metrics are provided below

$$OA = \frac{\sum_{k=1}^{N} TP_k}{\sum_{k=1}^{N} TP_k + FP_k + TN_k + FN_k} \tag{30}$$

$$MIoU = \frac{1}{N} \sum_{k=1}^{N} \frac{TP_k}{TP_k + FP_k + FN_k} \tag{31}$$

$$precision_k = \frac{TP_k}{TP_k + FP_k} \tag{32}$$

$$Recall_k = \frac{TP_k}{TP_k + FN_k} \tag{33}$$

$$F1Score = \frac{2 \times (precision_k \times Recall_k)}{precision_k + Recall_k} \tag{34}$$

where TP_k, FP_k, FN_k, and TN_k indicate the true positive, false positive, true negative, and false negatives, respectively, for object indexed as class k.

D. Quantitative Comparison Diverse Methods

This section evaluates the performance of the proposed semantic segmentation method by comparing CGSNet with various state-of-the-art methods. The compared methods include supervised models, self-supervised models, and fully unsupervised models. The supervised semantic segmentation methods include U-Net [35] and DeepLabv3+ [36], as well as advanced remote sensing image segmentation methods utilizing Transformers and frequency-domain analysis for global feature extraction, such as MANet [29], GCDNet [30], UNetFormer [37], CMTFNet [38], RS^3 Lambda [39], FSSFFormer [43], and HAGNet [44]. The self-supervised methods include AlignSeg [40], SDCluster [41], and SOD [42]. These self-supervised methods were uniformly trained using 10% of the labeled training set data. The fully unsupervised methods include SSAC [45] and ATSG [46], both of which do not use any annotated data.

UNetFormer employs a lightweight ResNet18 encoder and a transformer-based decoder to establish long-range dependencies, thereby enhancing the utilization efficiency of global and local information. RS^3 Lambda proposes a novel dual-branch network combining visual state space blocks with CNNs and introduces a collaborative completion module to effectively integrate global and local features, achieving excellent remote sensing image segmentation performance while maintaining computational efficiency. FSSFFormer leverages multiresolution wavelet decomposition to explicitly preserve global spatial structures while retaining fine-scale details, employing spectral enhancement and spectral–spatial context attention modules to enhance boundary fidelity and contextual modeling. HAGNet introduces a multiresidual encoder–decoder with hierarchical attention gates and a mixture-of-head attention module for bidirectional semantic interaction, while a hierarchical interaction loss enforces cross-level consistency, effectively mitigating prediction inconsistencies in complex landscapes. AlignSeg proposes an end-to-end self-supervised segmentation paradigm, addressing semantic inconsistency in Transformer-based dense representations through prototype-image alignment and global–local alignment with attention constraints, thereby improving segmentation accuracy in non-salient regions while enabling real-time inference. SDCluster proposes a pixel-level clustering SSL method, extending the hierarchical self-detection framework to the pixel level and incorporating a clustering assignment module and a clustering constraint module containing prototype bundles and semantic consistency constraints. This method not only allows qualitative assessment through visualized clustering results but

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TABLE I
EXPERIMENT RESULTS ON THE ZURICH (VEGETATION), ZURICH (WATER BODY), GID (WATER BODY), AND GID (VEGETATION) THE BOLD VALUES INDICATE THE BEST RESULTS

Methods	Supervision method	GID (Vegetation)			Zurich (Vegetation)			GID (Water body)			Zurich (Water body)		
		F1 (%)	OA (%)	mIoU (%)	F1 (%)	OA (%)	mIoU (%)	F1 (%)	OA (%)	mIoU (%)	F1 (%)	OA (%)	mIoU (%)
U-Net [35]	Supervision	66.21	62.88	45.58	60.90	71.44	53.53	85.73	86.81	78.29	90.45	94.24	87.32
DeepLabv3+ [36]		66.94	70.08	53.78	61.93	72.37	54.61	86.89	87.73	79.03	91.75	92.75	90.01
MANet [29]		70.09	67.66	50.94	65.29	74.07	55.18	87.76	88.33	80.69	93.93	97.22	92.48
GCDNet [30]		68.39	69.42	53.05	66.06	67.44	56.32	89.08	90.46	81.21	94.21	96.74	92.31
UNetFormer[37]		73.02	69.49	57.51	65.28	72.22	55.43	89.25	90.66	81.70	94.34	97.27	92.88
CMTFNet[38]		77.83	62.85	62.83	70.23	75.24	59.57	91.42	93.78	86.35	95.12	97.11	93.33
RS3Mamba[39]		79.36	73.85	65.78	74.29	78.92	63.16	92.73	94.55	91.52	96.10	98.31	95.20
FSSFFormer[43]		80.13	74.42	65.94	75.16	79.58	63.70	93.09	94.83	91.62	96.33	98.45	95.28
HAGNet[44]		81.27	75.09	66.24	76.03	80.14	64.51	**93.36**	**95.01**	**91.86**	**96.58**	**98.63**	**95.40**
SSAC [45]	Unsupervised	58.65	63.08	45.19	58.62	68.94	50.83	66.72	77.06	67.58	85.05	93.17.	86.18
ATSG [46]		60.47	64.72	49.24	60.23	70.25	51.63	67.31	77.58	67.91	86.05	94.62	86.93
AlignSeg[40]	Self-supervision	52.79	65.12	46.27	55.84	67.36	49.36	82.27	83.43	75.92	86.74	93.78	85.15
SDCluster[41]		57.12	62.32	44.48	57.93	68.29	50.07	65.53	76.66	66.81	84.65	94.03	85.48
SOD[42]		64.38	68.48	59.69	68.68	70.71	54.57	84.22	85.31	76.75	88.70	94.10	86.01
CGSNet (Ours)	Self-supervision	**90.85**	**88.54**	**78.46**	**90.90**	**89.05**	**79.58**	89.19	92.25	87.83	94.92	95.32	93.64

also addresses the difficulty in evaluating prediction quality in current self-supervised methods, thereby enhancing fine-tuning accuracy and sample segmentation capability for semantic segmentation. SOD proposes a multimodal prior knowledge-guided self-supervised model. Through auxiliary tasks such as appearance representation learning, depth-enhanced SSL, and contrastive multimodal representation learning, it addresses the limited discriminability of multimodal representations, learns multimodal representations from limited labeled data, enhances intermodal compactness, and improves the performance of multimodal segmentation networks. SSAC introduces style-shift-aware category contrastive learning to explore domain-invariant information and compensate for style randomization-induced distortions, paired with a dual-branch augmentation strategy to enrich style diversity. ATSG employs an adaptive token linking module to analyze semantic diversity in ViTs and a SAM-guided boundary refiner that incorporates object and boundary priors, effectively improving pseudo-label quality and small object segmentation completeness.

As shown in Table I, the numerical results on the Zurich (vegetation), Zurich (water), GID (water), and GID (vegetation) datasets demonstrate that CGSNet outperforms other networks by a significant margin in terms of $F1$ score, OA, and mIoU. Specifically, on the Zurich (vegetation) dataset, CGSNet achieved the highest accuracy with an $F1$ score of 90.90%, OA of 89.05%, and mIoU of 79.58%. On the Zurich (water body) dataset, CGSNet achieved an $F1$ score of 94.92%, OA of 95.32%, and mIoU of 93.64%. On the GID (vegetation) dataset, CGSNet achieved an $F1$ score of 90.85%, OA of 88.54%, and mIoU of 78.46%. Across all evaluation metrics, CGSNet achieved the highest segmentation accuracy in different semantic segmentation tasks.

Fig. 3 shows a comparison of vegetation detection results between CGSNet and the index-based SSL method (SOD). In terms of segmentation performance, CGSNet demonstrates strong capability in identifying easily confused targets like bare land and dry areas, effectively segments small targets, and achieves excellent segmentation results for complex target edges and contours.

Fig. 4 presents the experimental results of water body detection by CGSNet compared with segmentation performance. CGSNet exhibits a strong ability to distinguish water bodies from easily confused objects like vegetation and buildings, while maintaining high detection accuracy for water bodies of different colors. Furthermore, the proposed method achieves outstanding segmentation results for water bodies and their complex boundary regions.

As an advanced shadow detection method, the SDI constructs a transformation model by analyzing the spectral differences between the NIR and visible light bands to suppress interference from vegetation and water bodies. By integrating the normalized CDCI and the CPI, SDI achieves relatively high-precision shadow detection [20]. However, such physics model-based index methods primarily rely on pixel-level spectral calculations, lack understanding of image semantic information, and struggle to effectively model contextual relationships between pixels. Consequently, SDI often exhibits significant false detections and missed detections in complex scenes. To address these limitations, we utilized SDI as prior information and conducted shadow detection experiments with CGSNet on the GF-2 dataset. The experimental results are shown in Fig. 5. Since SDI heavily depends on pixel grayscale calculations, dark water bodies are frequently misclassified as shadows (as highlighted by the red boxes). The experimental results demonstrate that CGSNet can effectively leverage the prior information provided by SDI, adaptively learning and extracting meaningful shadow features, thereby significantly overcoming SDI's shortcomings of high false detection rates and weak anti-interference capability. Ultimately, CGSNet achieves more accurate and robust shadow detection results.

E. Generalization Ability

Multispectral images possess rich texture and semantic features; object recognition algorithms based on multispectral

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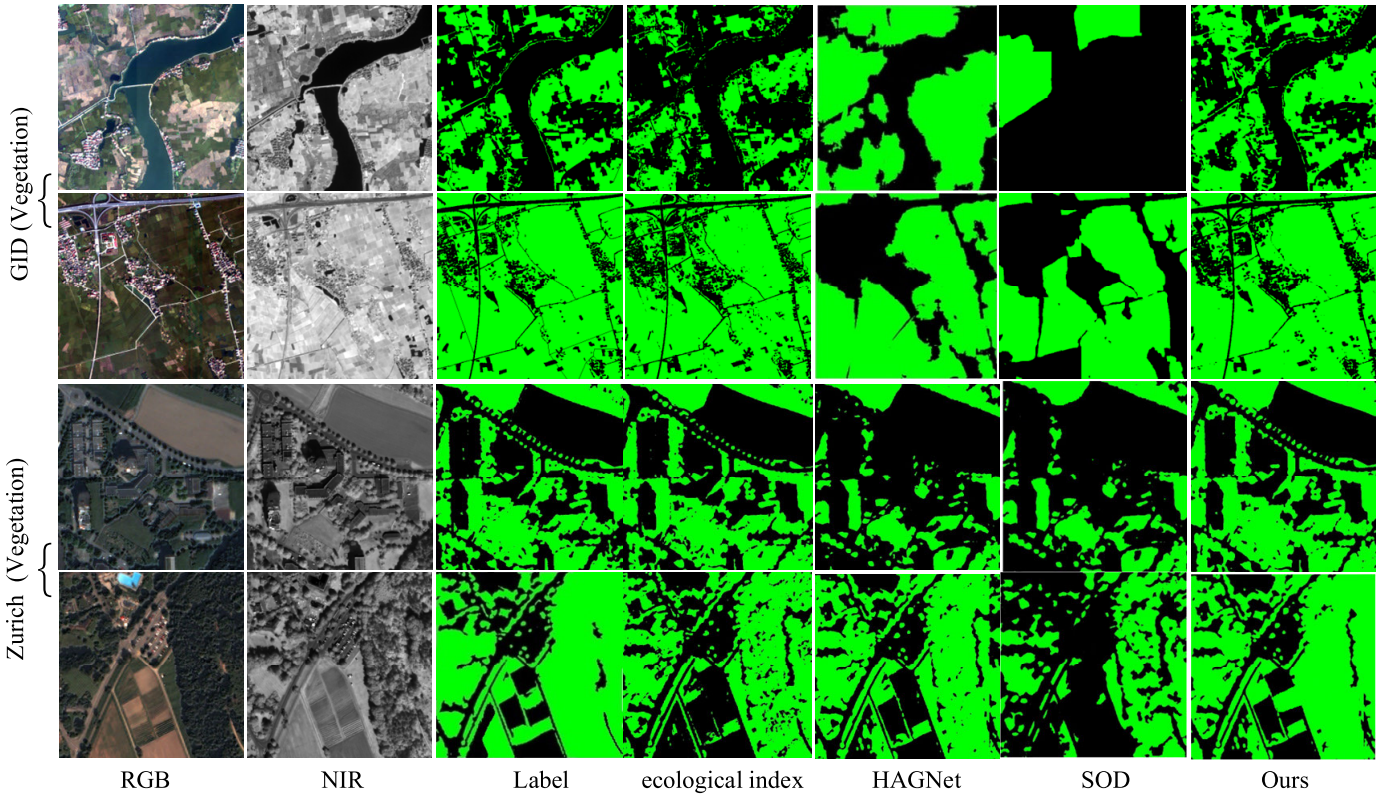


Fig. 3. Complete results of different methods. The first two rows are the test results of the GID (vegetation) dataset. The last two rows are the test results of the Zurich (vegetation) dataset.

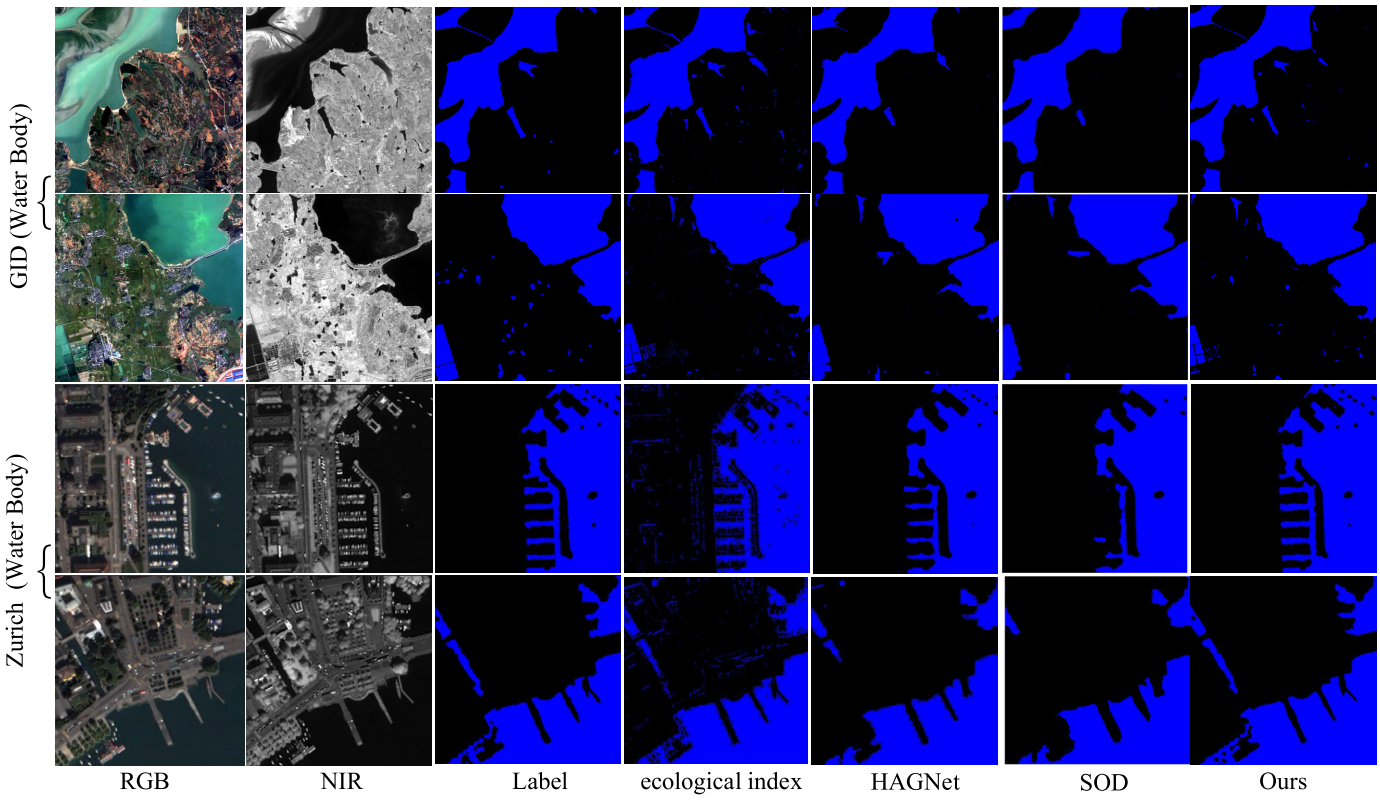


Fig. 4. Complete results of different methods. The first two rows are the test results of the GID (water body) dataset. The last two rows are the test results of the Zurich (water body) dataset.

images are susceptible to influences from different scenes and environmental factors, often lacking cross-dataset generalization capability. In contrast, multimodal prior data typically focus on the fundamental physical attributes of

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Fig. 5. Comparison of shadow detection results between CGSNet and SDI.

TABLE II

QUANTITATIVE COMPARISON OF ZERO-SAMPLE VEGETATION AND WATER BODY DETECTION RESULTS ON ZURICH (VEGETATION) AND ZURICH (WATER BODY) DATASETS USING THE MODEL TRAINED ON GID (VEGETATION) AND GID (WATER BODY) DATASETS, RESPECTIVELY

Methods	Supervision method	Trained on GID (Vegetation) →Zurich (Vegetation)			Trained on GID (Water body) →Zurich (Water body)		
		F1 (%)	OA (%)	mIoU (%)	F1 (%)	OA (%)	mIoU (%)
DeepLabv3+	Supervision	26.57	59.43	35.29	78.53	82.42	76.81
MANet		35.18	60.75	38.83	81.68	85.97	80.80
GCDNet		36.77	61.43	39.67	83.52	86.73	82.55
UNetFormer		48.14	70.08	48.86	84.08	88.27	83.68
CMTFNet		53.56	72.07	54.55	84.14	89.55	83.92
RS^{3}Mamba		56.75	72.84	54.39	85.51	90.64	85.53
FSSFFormer		55.81	72.49	53.90	85.28	90.37	85.48
HAGNet		56.93	73.24	54.87	85.83	91.14	85.87
SSAC	unsupervised	48.03	58.12	40.38	81.12	88.86.	81.73
ATSG		50.57	60.49	42.63	82.54	89.80	82.31
SOD	Self-supervision	49.77	60.86	43.94	82.92	87.05	80.36
AlignSeg		50.43	61.52	45.07	84.32	90.08	82.35
CGSNet	Self-supervision	**89.80**	**88.16**	**78.21**	**90.73**	**95.83**	**90.69**

objects and are less affected by varying scenes and environments. Therefore, effectively learning features from prior data can significantly enhance a model's cross-dataset generalization capability.

To validate the generalization capability of CGSNet across different datasets, we trained CGSNet on the GID (vegetation) and GID (water) datasets, respectively. Subsequently, the trained models were directly used for vegetation and water body detection experiments on the Zurich (vegetation) and Zurich (water) datasets without any fine-tuning or retraining. The test results and quantitative metrics are shown in Fig. 6 and Table II. Fig. 6 shows that CGSNet can effectively detect vegetation and water bodies on the Zurich dataset under these sampling conditions. Similarly, we trained CGSNet on the Zurich (vegetation) and Zurich (water) datasets, respectively, and then used the trained models for vegetation and water body detection experiments on the GID (vegetation) and GID (water) datasets. The test results and quantitative metrics are shown in Fig. 7 and Table III.

TABLE III

QUANTITATIVE COMPARISON OF ZERO-SAMPLE VEGETATION AND WATER BODY DETECTION RESULTS ON GID (VEGETATION) AND GID (WATER BODY) DATASETS USING THE MODEL TRAINED ON ZURICH (VEGETATION) AND ZURICH (WATER BODY) DATASETS, RESPECTIVELY

Methods	Supervision method	Trained on Zurich (Vegetation) →GID(Vegetation)			Trained on Zurich (Water body) →GID(Water body)		
		F1 (%)	OA (%)	mIoU (%)	F1 (%)	OA (%)	mIoU (%)
DeepLabv3+	Supervision	49.72	65.28	47.34	51.88	85.93	60.42
MANet		54.37	65.97	47.80	52.03	86.10	60.82
GCDNet		56.08	66.38	48.03	52.16	86.43	61.07
UNetFormer		58.58	66.43	48.85	55.28	87.40	63.22
CMTFNet		60.73	67.60	50.39	68.39	88.93	72.64
RS^{3}Mamba		65.28	69.42	51.18	73.42	90.28	76.85
FSSFFormer		63.43	68.51	50.75	71.36	89.15	74.29
HAGNet		66.17	70.32	51.48	72.25	89.63	74.51
SSAC	unsupervised	47.49	61.04	44.63	70.94	90.28	74.60
ATSG		49.17	63.28	45.96	73.78	89.53	76.81
SOD	Self-supervision	48.07	61.86	45.37	71.28	89.65	74.87
AlignSeg		50.68	64.39	46.81	74.53	90.32	77.45
CGSNet	Self-supervision	**90.68**	**88.27**	**77.56**	**90.51**	**95.43**	**89.18**

TABLE IV

PERFORMANCE COMPARISON WITH DIFFERENT τ VALUES ON THE GID (VEGETATION) DATASET

τ	mIoU (%)	OA (%)	F1 (%)
0.1	77.52	87.86	90.08
0.2	**78.46**	**88.54**	**90.85**
0.3	77.84	88.05	90.36
0.4	76.91	87.43	89.52

TABLE V

PERFORMANCE COMPARISON OF THRESHOLDING SCHEMES ON THE GID (VEGETATION) DATASET

Schemes	mIoU (%)	OA (%)	F1 (%)
Conventional Otsu	76.83	86.12	88.45
ATS	**78.46**	**88.54**	**90.85**

Experimental results demonstrate that CGSNet outperforms other models in terms of its ability to effectively extract intricate image features, accurately discern target locations and boundaries, and exhibit superior generalization $\tau\tau$ capabilities. Moreover, it addresses the issue of inadequate model robustness in semantic segmentation tasks. Our approach exhibits excellent performance on diverse datasets and achieves superior segmentation outcomes compared to alternative networks.

F. Ablation Study

Ablation studies were conducted to assess the effectiveness of key components in the proposed CGSNet. Tables IV–VIII provide a detailed account of these experiments and their

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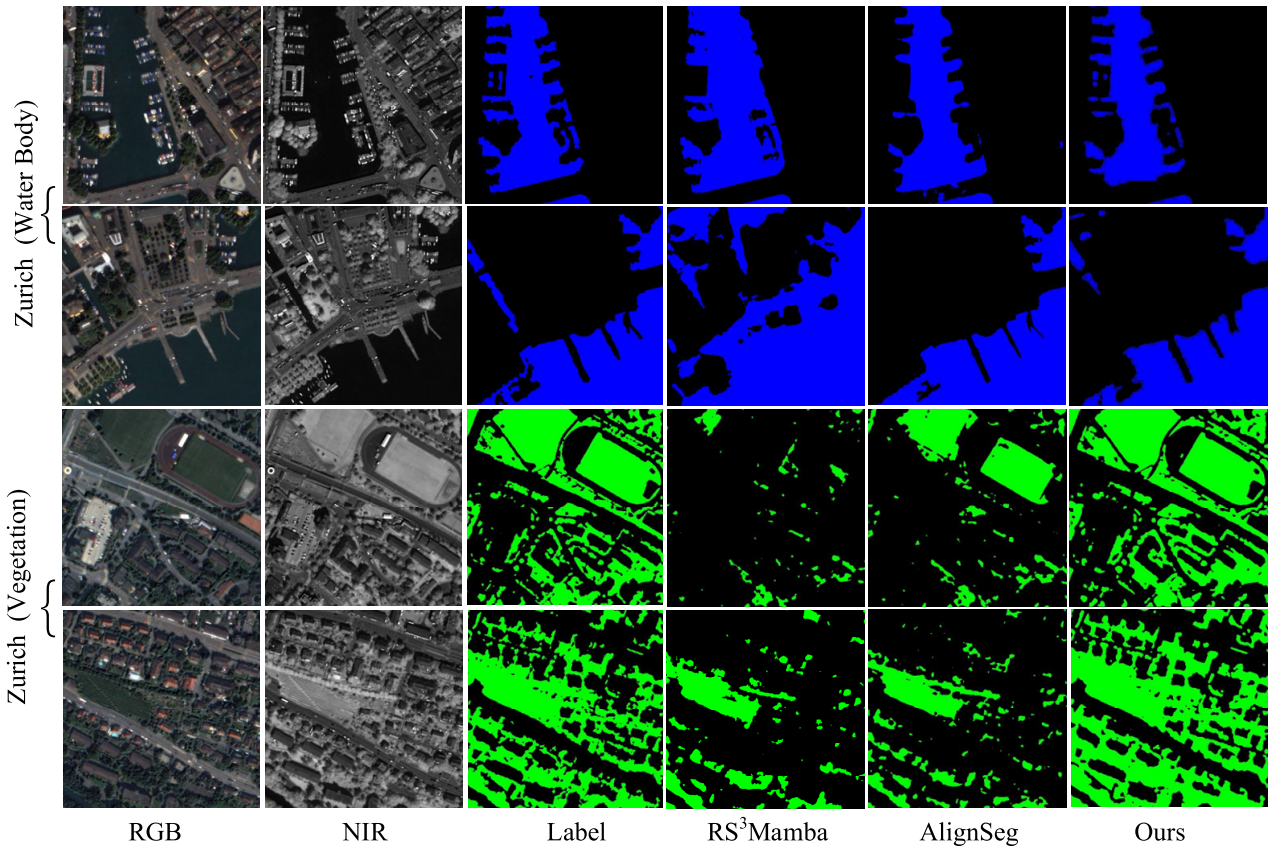


Fig. 6. Results of zero-shot detection on Zurich (vegetation) and Zurich (water body) datasets using the models trained on GID (vegetation) and GID (water body) datasets, respectively.

TABLE VI

COMPARISON OF VEGETATION AND WATER BODY DETECTION RESULTS ON THE GID AND ZURICH DATASETS USING MODELS TRAINED WITH DIFFERENT LABELS ON THE GID (VEGETATION) AND GID (WATER BODY) DATASETS

Labels	Trained on GID (Vegetation)						Trained on GID (Water body)					
	→ GID(Vegetation)			→ Zurich(Vegetation)			→ GID(Water body)			→ Zurich(Water body)		
	F1(%)	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)
NDVI	88.26	86.08	77.41	87.59	86.36	76.60	-	-	-	-	-	-
NDWI	-	-	-	-	-	-	88.43	90.94	86.07	89.58	93.88	89.08
GL	80.32	75.98	68.58	57.32	73.64	55.78	87.95	90.73	85.92	84.21	88.68	83.37
OT	86.79	85.80	76.92	85.76	84.61	75.83	87.23	90.60	85.64	89.13	93.56	88.71
LV	89.73	87.52	77.68	88.15	87.06	77.28	88.59	91.17	86.28	89.65	94.10	89.27
IV	90.08	87.83	78.25	89.23	87.59	77.63	89.03	91.84	86.96	90.17	94.62	89.80
APL	**90.85**	**88.54**	**78.46**	**89.80**	**88.16**	**78.21**	**89.19**	**92.25**	**87.83**	**90.73**	**95.83**	**90.69**

TABLE VII

ABLATION EXPERIMENT RESULTS CONSIDERING DIFFERENT MODULES OF APL

APL			GID (Vegetation)		GID (Water body)	
MFG	MO	PMG	OA (%)	mIoU (%)	OA (%)	mIoU (%)
	√	√	86.92	77.25	91.14	86.75
√		√	88.27	78.33	92.03	87.59
√	√		87.95	77.41	91.02	87.14
√	√	√	**88.54**	**78.46**	**92.25**	**87.83**

TABLE VIII

ABLATION EXPERIMENT RESULTS CONSIDERING DIFFERENT MODULES OF CGSNET

MCSM	MEO	GID (Vegetation)		GID (Water body)	
		OA(%)	mIoU(%)	OA(%)	mIoU(%)
		86.67	77.37	90.43	85.92
√		88.15	78.28	91.49	87.34
	√	87.45	77.80	91.08	86.81
√	√	**88.54**	**78.46**	**92.25**	**87.83**

results. The symbol indicates that the corresponding module was retained, while the absence of this symbol indicates that the module was removed. All other settings, including loss functions and optimizers, remained consistent with the complete CGSNet.

1) Effectiveness of the ATS: To rigorously evaluate the ATS component, we conducted additional ablation experiments. First, we analyzed the sensitivity of the transition width τ

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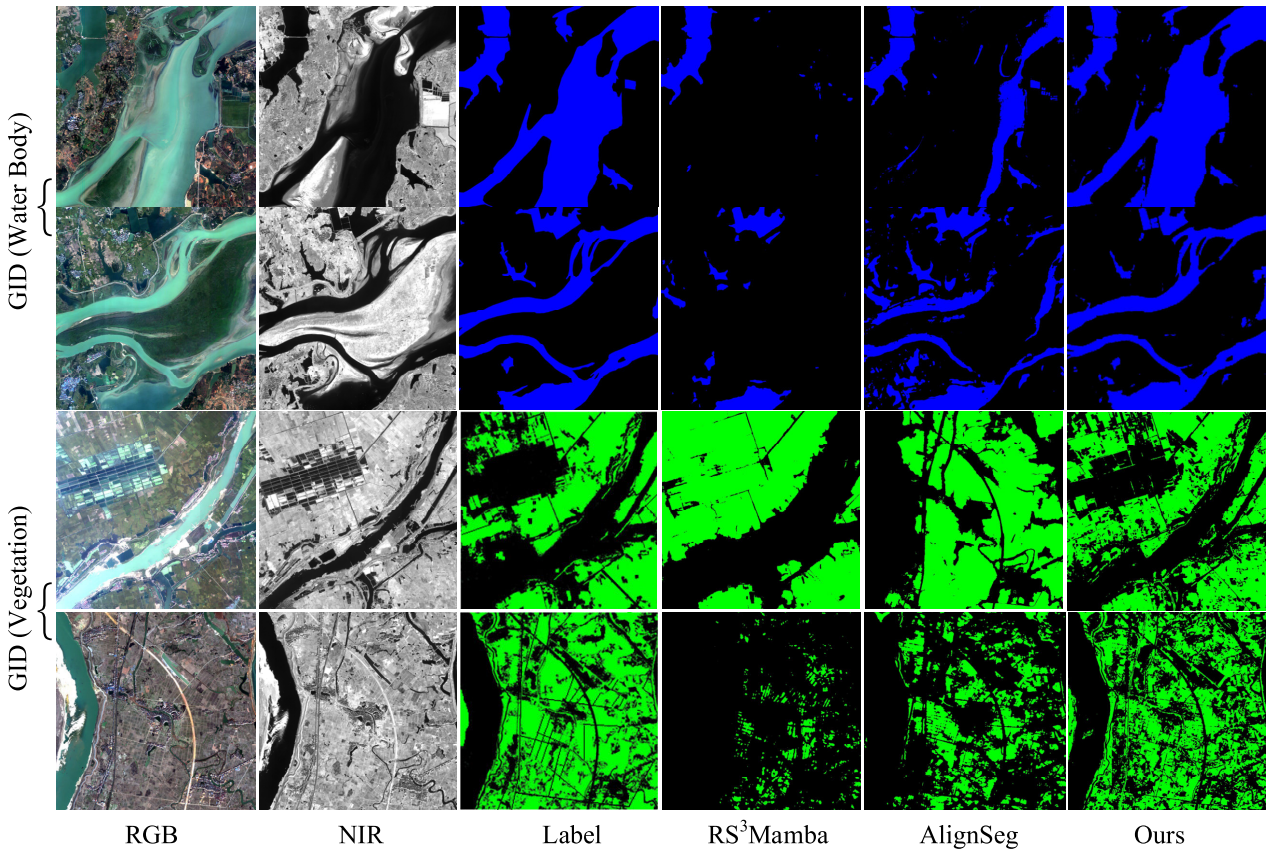


Fig. 7. Results of zero-shot detection on GID (vegetation) and GID (water body) datasets using the models trained on Zurich (vegetation) and Zurich (water body) datasets, respectively.

on the GID (vegetation) dataset. As shown in Table IV, the model achieves optimal performance at $\tau = 0.2$. A smaller τ fails to adequately model boundary transitions, whereas a larger τ introduces noise from over-expanded transition regions. Second, since the weighting coefficient β is adaptively derived from the skewness coefficient rather than a tunable hyperparameter, we validated its effectiveness by comparing our ATS against conventional Otsu thresholding (OT) (without skewness weighting). As reported in Table V, our ATS outperforms conventional Otsu by 1.63% in mIoU, confirming that adaptively weighting classes according to histogram asymmetry effectively corrects threshold bias caused by imbalanced foreground-background distributions in category prior images.

2) Effectiveness of the Pseudo-Label Generation Strategy: We removed the APL method and instead used the vegetation annotation masks provided by the GID dataset as ground truth labels (GLs) to train the model. The vegetation detection performance was then evaluated on the GID and Zurich datasets, using the finely manually annotated vegetation labels to compute quantitative metrics. The test results in Table VI show that employing the APL mechanism effectively improves the model's detection accuracy and cross-dataset generalization capability.

Fig. 8 shows a visual comparison of vegetation detection results on the GID (vegetation) dataset when using GL and APL as labels. The results indicate that using APL significantly enhances the model's ability to represent details of land cover targets. Utilizing spectral indices to generate pseudo-labels effectively improves the model's capability to detect complex targets and its generalization across datasets. We compared several pseudo-label generation methods, including the following.

1) *Soft Label Baseline (NDVI/NDWI):* Directly using the raw normalized index values as per-pixel probability labels without any thresholding or confidence weighting.
2) *Otsu Thresholding:* Applying the Otsu threshold algorithm to ecological indices to obtain binary masks as training pseudo-labels.
3) *Linear Variation:* Generating pseudo-labels using dynamic thresholds that increase linearly with the number of epochs to segment the ecological indices. Specifically, over the total number of epochs N, the ecological index value a is considered from 0 to 255. The linear variation (LV) threshold is calculated as

$$T_L = T_{low} + \left(T_{high} - T_{low}\right) \times \frac{n}{N} \tag{35}$$

where T_{low} represents the minimum threshold (set to 20), T_{high} denotes the maximum threshold (set to 255), n indicates the current epoch, and N corresponds to the total number of epochs (set to 100).
4) *Index Variation:* Using a dynamic threshold that increases exponentially with the epoch number to segment ecological indices and generate pseudo-labels. The

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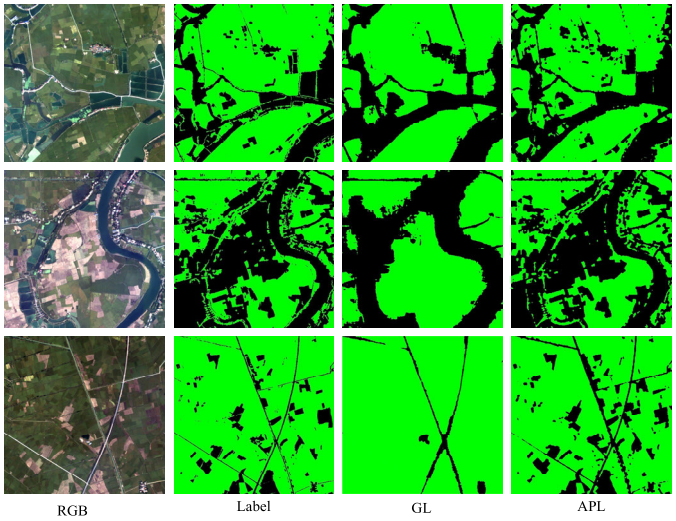


Fig. 8. Vegetation detection results on the GID (vegetation) dataset using GL versus APL as training labels.

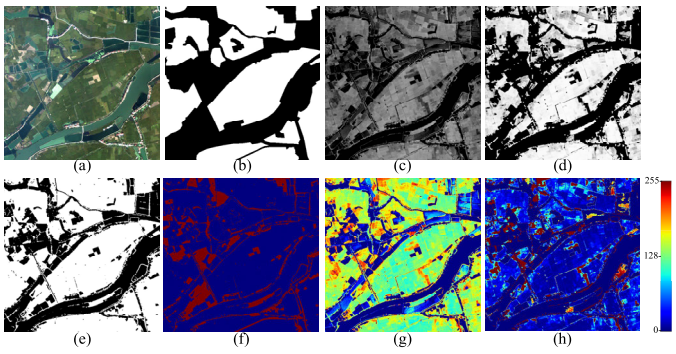


Fig. 9. Discrepancy heatmaps between different labels and the manually finely annotated vegetation label. (a) RGB image. (b) Vegetation label of original GID dataset. (c) NDVI. (d) APL label. (e) Manually finely annotated label. (f) Discrepancy heatmap between the original GID label and the manually finely annotated label. (g) Discrepancy heatmap between NDVI and the manually finely annotated label. (h) Discrepancy heatmap between the APL label and the manually finely annotated label.

exponential index variation (IV) is calculated using the formula

$$T_E = T_{low} + \left(T_{high} - T_{low}\right) \times \left(1 - e^{-\frac{n}{N}}\right). \tag{36}$$

Experimental results comparing these different pseudo-label generation methods are summarized in Table VI. The findings confirm that the adaptive confidence-weighted label generation mechanism achieves the optimal detection accuracy and exhibits the strongest generalization capability.

To more intuitively demonstrate the quality of labels generated by the adaptive confidence weighting method, we compared the GL label, NDVI label, and APL label with the manually finely annotated vegetation label. Fig. 9 presents the discrepancy heatmaps between these labels and the manually finely annotated vegetation label, where larger pixel-wise differences are indicated by darker red hues. The results indicate that the labels produced by the adaptive confidence weighting method exhibit the closest agreement with the manual fine annotations, demonstrating the effectiveness of the proposed approach.

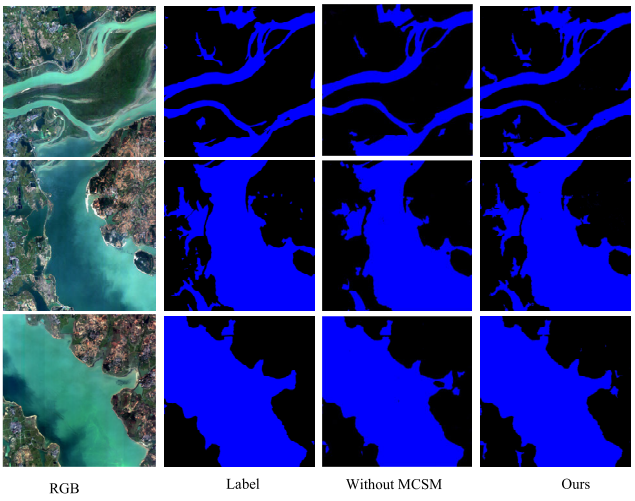


Fig. 10. Semantic segmentation comparison for CGSNet and CGSNet without MCSM on the GID (water body) dataset.

Furthermore, we validated the effectiveness of individual steps within the APL method.

1) *APL Without MFG:* We removed the guidance module using RGB features for label generation, retaining only the pseudo-label generation method with a smoothed probability distribution. As shown in Table VII, incorporating multispectral feature guidance (MFG) effectively improves the model's semantic segmentation performance. This indicates that multispectral image features can effectively guide the model to correct erroneous parts in the category prior images.
2) *APL Without MO:* To verify the effectiveness of incorporating morphological optimization (MO) in the APL module, we tested the model performance without MO. As shown in Table VII, incorporating MO effectively enhances the model's semantic segmentation performance.
3) *APL Without PMG:* To verify the effectiveness of the PMG module, we tested the model performance without this module. As shown in Table VII, incorporating PMG effectively improves the model's semantic segmentation performance, indicating that this strategy can effectively learn cross-modal invariant features and enhance the training effectiveness of the self-supervised model.

3) Effectiveness of the Multimodal CGSM: To validate the effectiveness of the cross-gated self-attention module (MCSM), we replaced it with standard multilayer convolutions for feature extraction and conducted comparative experiments with this modified model. The experimental results (Table VIII and Fig. 10) show that the cross-gated self-attention module significantly reduces the false detection rate in easily confused regions.

4) Effectiveness of the MEO: To validate the effectiveness of the MEO module, we tested a model that solely relies on the cross-entropy loss function l_{seg} for calculating the loss of model predictions, without employing the edge optimization loss function l_{edge}. Table VIII demonstrates after adding the

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TABLE IX
COMPUTATIONAL COMPLEXITY ANALYSIS FOR MEASURING 512 × 512 IMAGES ON A SINGLE NVIDIA GEFORCE RTX 3090TI GPU. THE MIOU VALUE IS THE RESULT ON THE GID (VEGETATION) DATASETS

Methods	Baseline	FLOPs (G) ↓	Parameter (M) ↓	mIoU (%) ↑
U-Net	U-Net	394.58	27.91	45.58
Deeplabv3+	ResNet-101	138.63	40.35	53.78
UNetFormer	ResNet-50	46.04	<u>24.20</u>	57.51
MANet	ResNet-50	283.74	38.45	50.94
CMTFNet	ResNet-50	<u>32.85</u>	30.07	62.83
FSSFFormer	U-Net	48.27	58.50	65.94
HAGNet	U-Net	65.98	**18.71**	<u>66.24</u>
CGSNet (Ours)	Mobilenetv2	**17.82**	42.48	**78.46**

MSB module to the baseline network, the performance of OA on the GID (vegetation) is improved by 0.39%.

G. Generalization Ability

We evaluate the computational complexity of the proposed CGSNet using the following evaluation metrics: the floating-point operation count (FLOP), the number of model parameters, and the memory footprint. We fed input data of size 512 × 512 into all models and also evaluated the computational complexity and parameter sizes of various methods in the same runtime environment, and the results are shown in Table IX. From Table IX, CGSNet has 17.82-G FLOPs and 42.48-M parameters, which is less than the model complexity of other methods, and achieves the optimal segmentation accuracy, showing the superior performance of CGSNet.

V. CONCLUSION

This article proposes a category prior-guided self-supervised network for remote sensing semantic segmentation. First, an APL mechanism is proposed that leverages multispectral images and their derived category prior information to produce high-quality pseudo-labels. Second, a multimodal cross-gated self-attention module (MCSM) is designed to achieve complementary fusion of cross-modal features. Third, the MEO loss enhances boundary coherence across different resolutions without additional computational cost. By integrating domain-specific spectral knowledge into self-supervision, CGSNet overcomes the limitations of dependency on external supervision and underutilized spectral physics in existing methods. Extensive experiments demonstrate CGSNet's superior generalization capability and robustness, particularly in cross-dataset evaluations. Ablation studies further validate the contributions of each component. This work advances self-supervised segmentation by intrinsically encoding spectral-domain priors, providing a practical solution for land-cover classification with unlabeled data.

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